

Description

The **ProMicro NRF52840** is a micro development board designed as a Pro Micro alternative. It is compatible with the Nice!Nano and has the **same pin layout** as the ProMicro. This allows it to be used with almost all ProMicro keyboards. The ProMicro NRF52840 development board comes with a lithium battery charging and discharging circuit that supports 3.7V lithium battery charging and discharging, with a charging rate of 100mA per hour. It also comes with a software-controlled switch that can cut off the power supply to the LED to reduce power consumption. In the off state, its standby current can be reduced to 1mA.

This development board is equipped with the **nRF52840** wireless system-on-chip (SoC), which has **high performance** and **low power consumption**. The nRF52840 uses an **ARM Cortex-M4F** processor that runs at up to **64MHz** and is equipped with **1MB** flash memory and **256KB** RAM. It supports **multiple wireless protocols** including **Bluetooth 5.0, Thread, Zigbee, ANT and 2.4GHz**. In addition, the chip also integrates **various peripheral interfaces** such as ADC, PWM, SPI, I2C, UART, USB and GPIO. In terms of security performance, nRF52840 provides **multiple security features** including AES encryption, SHA-256 hash algorithm and true random number generator (TRNG).

Note: The default factory built-in program is Blink-All-IO (all IO ports automatically flip in 1S). If you want to enter the Nice! Nano V2 Bootloader, please short RST to GND twice within 0.5S. The keyboard board can be directly pressed twice within 0.5S.

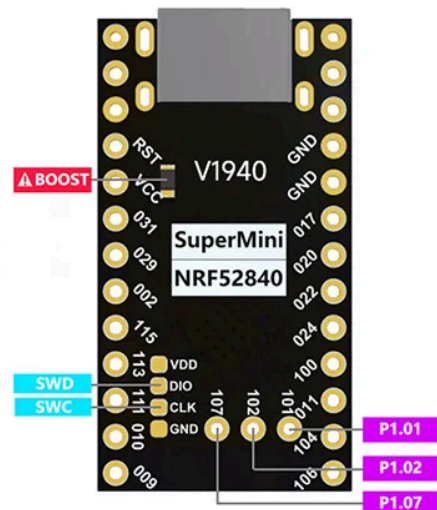
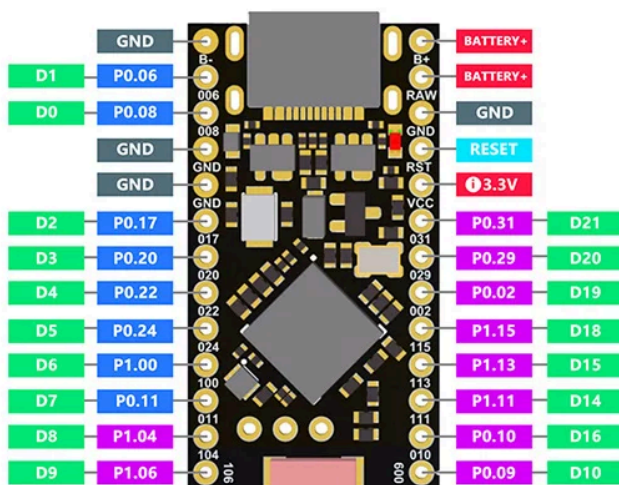
Feature

- Adopt NRF52840 chip, ARMCortex-M4F processor, main frequency of 64MHz;
- Built-in 1MB flash memory and 256KB RAM;
- Onboard antenna, supports multiple wireless protocols such as Bluetooth 5, Thread, Zigbee, ANT and 2.4GHz;
- LTH7R Battery charging IC supports charging and discharging of lithium batteries;
- Power consumption is only 20uA when in standby mode;
- Multiple peripheral interfaces such as ADC, PWM, SPI, I2C, UART, USB and GPIO, etc.;
- Supports multiple security protocols including AES encryption, SHA-256 hash algorithm and true random number generator (TRNG);



Charge Current Select Jumper:
 Connect this jumper "BOOST", the charging current from 100mA to 300mA.
 Do this only if your battery capacity is larger than 500mAh to avoid explosions.

External VCC Cutoff Control:
 When P0.13 is set low, the power supply to the 3.3V-VCC pin will be turned off.
 This is useful for components that use less power when idle (such as RGB LEDs).



Specification

CPU	NRF52840 Chip with ARMCortex-M4F processor
Frequency	64MHz
Flash Memory	Built-in 1MB
RAM	256KB
Wireless Protocols	Support Bluetooth 5, Thread, Zigbee, ANT and 2.4GHz
Interface	ADC、PWM、SPI、I2C、UART、USB and GPIO
Supply Voltage	5v
Battery Interface	3.7v
Compatibility	Compatible with nice! Nano V2.0 and Promicro
Operating Temperature	0°C to 50°C
Dimensions	17.8*33mm